Ziqian Guo

China Agricultural University Mechanical Design, Manufacturing and Automation Class 242

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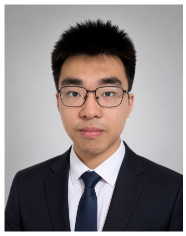

Focus: Humanoid Robot Soccer, Motion Control, Reinforcement Learning, ROS2 Strategy and Deployment

Education

China Agricultural University Sept. 2024 – Present

B.Eng. in Mechanical Design, Manufacturing and Automation *Beijing, China*

- English: CET-4 593, CET-6 554; able to read English papers, open-source documentation, and robotics technical materials.
- Relevant coursework: Engineering Mechanics, Fundamentals of Circuits and Electronics, Control Engineering, Data Structures, Probability and Statistics, College Physics.
- Academic focus: humanoid robot soccer strategy, legged robot motion control, reinforcement learning, and simulation-to-real deployment.

Competition Experience

CAU RoboCup Humanoid Robot Team Oct. 2025 – Present

Core Member / Strategy and Locomotion Deployment *China Agricultural University*

- Participated as a core team member in humanoid robot soccer competitions, working on match strategy, ROS2 communication, behavior switching, policy deployment, and hardware debugging.
- Champion, RoboCup China Middle Group; Champion, RoboLeague National Championship Wuhan Division; Runner-up, RoboCup World Championship Incheon Large Group.
- Champion, National University Robot Soccer Super League Beijing Monthly Competition; participated in pre-match testing, field adaptation, strategy tuning, and troubleshooting.
- Contributed to robot soccer strategy design, including role assignment, offense-defense switching, kick-off/set-play handling, and game-state response.
- Familiar with basic ROS2 communication workflow, including topic, node, launch, parameter configuration, and module integration; basic development experience in C++ and Python.

Open-source Projects

booster_mjlab: Reinforcement Learning Locomotion Training for Booster T1/K1 May 2026 – Present

Personal Repository / Developer

Project link: github.com/qianxvde/booster_mjlab

- Built an MJLab-based locomotion training project for Booster T1/K1 humanoid robots, including robot models, training tasks, motion data, and running scripts.
- Supports flat-ground and rough-terrain walking tasks for T1/K1, and can be used for training, testing, and replaying humanoid walking policies.
- Organized the training workflow, including environment registration, training scripts, policy testing, model replay, and training log inspection.
- The project serves RoboCup-related scenarios and provides a foundation for stable walking, reference motion tracking, and future kicking policy training.

hhtools: Humanoid Motion Retargeting and Data Conversion Toolkit June 2026 – Present

Personal Repository / Developer

Project link: github.com/qianxvde/hhtools

- Organized a motion data processing toolkit for Booster humanoid robots based on the human-humanoid-tools workflow.
- Supports motion segment clipping, motion preview, and data format conversion, including CSV, PKL, and BVH inputs.
- Added Booster-specific adapters so that external human motion data can be more easily used for T1/K1 motion imitation and reinforcement learning training.
- The project connects human motion data, robot motion data, and the training repository, preparing data for walking and kicking motion learning.

booster_wbc_fsm: Robot Policy Deployment and Safety FSM Framework July 2026 – Present

Personal Repository / Developer

Project link: github.com/qianxvde/booster_wbc_fsm

- Designed a lightweight finite-state-machine framework for Booster robot deployment, covering standing, walking, policy execution, and safety protection.

- Integrated trained locomotion policies into the deployment workflow and checked key configuration consistency, such as joint order, observation input, action scale, and control frequency.
- Added command filtering and safety-check logic to reduce risks during direct hardware deployment.
- Designed a staged debugging workflow, including passive check, damping protection, default standing, low-speed walking, and log replay for deployment troubleshooting.

Research Training

National Undergraduate Innovation Training Program Apr. 2026 – Present
Participant *Legged Robotics / Reinforcement Learning*

- Participated in technical planning, experiment implementation, and code repository management, focusing on reinforcement learning control and simulation training for legged robots.
- Combined paper reading, open-source framework reproduction, and robot platform experiments to build a workflow from motion data and simulation environments to policy deployment.

Skills

Programming	Python, basic C++, Linux / Ubuntu, Git, Shell, basic CMake, VS Code / Cursor
ROS2 & Deployment	Basic ROS2, topic / node / launch, parameter configuration, log debugging, module integration, robot policy deployment
Robot Strategy	RoboCup match strategy, role assignment, offense-defense switching, kick-off/set-play handling, game-state response
Simulation & Training	MuJoCo, MJLab, Isaac Sim, Isaac Lab, RSL-RL, PPO, AMP, simulation training and policy replay
Data Processing	AMASS, SMPL / SMPL-X, NPZ / PKL / BVH data processing, motion retargeting, robot joint mapping

Profile

- Long-term participant in humanoid robot soccer, with practical experience in match strategy, ROS2 integration, reinforcement-learning-based locomotion training, and robot deployment.
- Able to troubleshoot dependency, communication, simulation, training, and deployment issues in complex engineering environments.
- Currently focused on stable walking, kicking motion generation, match strategy decision-making, and low-level motion policy deployment for humanoid robot soccer.